

6.1 Introduction to Vectors

Scalars

Definition: Measurements that have *magnitude only*. Scalar variables are written $a, b, c, \dots$

Examples: distance, height, speed, mass, time, temperature, ...

Vectors

Definition: Measurements that have *magnitude and direction*. Vector variables are written $\vec{a}, \vec{b}, \vec{c}, \dots$

Examples: velocity (10 m/s [N]), acceleration (2 m/s² [NE]), force (50 N [fwd]), electrical currents (direction = along the wire), magnetic fields, momentum, electric fields, displacement, torque, ...

Two ways to describe a vector

- **Geometric:** drawn w.r.t. space/shape/orientation (e.g. N/E/S/W, or “moving forward”).
 - **Algebraic:** expressed using components on a standard Cartesian grid.
-

Notation & Magnitude

$\overrightarrow{AB}$ = vector from point A to point B. $-\overrightarrow{AB}$ = vector from B to A, i.e. $-\overrightarrow{AB} = \overrightarrow{BA}$.

Magnitude (size only):

$$|\vec{v}| \quad \text{or} \quad \|\vec{v}\|$$

Always: $|\vec{v}| = |-\vec{v}|$. If $\vec{v}$ is a vector, $\ominus \vec{v}$ has the same size but opposite direction.
⌚ “flips” direction

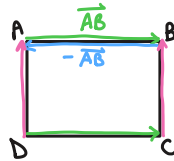
Equal Vectors & Location Independence

Key rule: Any vector with the same magnitude *and* direction is equal - **location does not matter**.

Rectangle ABCD:

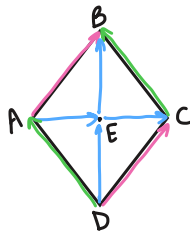
$$\overrightarrow{AB} = \overrightarrow{DC}, \quad \overrightarrow{AD} = \overrightarrow{BC}$$

But: $\overrightarrow{AB} \neq \overrightarrow{CD}$, $\overrightarrow{AC} \neq \overrightarrow{BD}$



Rhombus ABCD with centre E

$$\begin{aligned} \overrightarrow{AB} &= \overrightarrow{DC} \\ \overrightarrow{AE} &= \overrightarrow{EC} \\ -\overrightarrow{AD} &= \overrightarrow{CB} \\ |\overrightarrow{AB}| &= |\overrightarrow{AD}| = |\overrightarrow{BC}| = |\overrightarrow{DC}| \\ |\overrightarrow{BD}| &= |\overrightarrow{DB}| = 2|\overrightarrow{BE}| \end{aligned}$$



Vector Addition (Tip-to-Tail Rule)

To add two vectors, draw them **tip-to-tail**: $\vec{v} + \vec{u}$ is the vector from the tail of $\vec{v}$ to the tip of $\vec{u}$.



Examples:

$$\vec{AE} + \vec{EC} = \vec{AC}$$

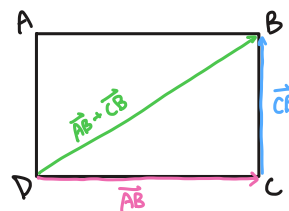
$$\vec{DA} + \vec{AB} = \vec{DB}$$

$$\vec{AB} + \vec{BC} = \vec{AC}$$

$$\vec{AB} + \vec{BD} = \vec{AD}$$

$$\vec{AB} + \vec{CB} = \vec{DB}$$

$$\vec{AD} + \vec{CB} = \vec{0} \leftarrow \text{zero vector}$$



Important: $|\vec{DA}| + |\vec{AB}| \neq |\vec{DB}|$ in general (triangle inequality; equality only when collinear).

Note: $\vec{0}$ = zero vector (magnitude 0, direction undefined); not the scalar 0.

The sum of two vectors is always a vector.

Basis & Algebraic Vectors

Basis = the directions that *define* a space. Algebraic vectors are given by their components along basis directions.

- **1D:** basis is the concept of "1"; every number on a number line is relative to it.
- **2D:** choose two directions that reach any point. Standard: x -direction $\vec{i} = (1, 0)$ and y -direction $\vec{j} = (0, 1)$ on the xy -grid.
- **3D:** uses $\vec{i}$, $\vec{j}$, and $\vec{k}$ directions.

Standard definitions: $\vec{i} = (1, 0)$, $\vec{j} = (0, 1)$.

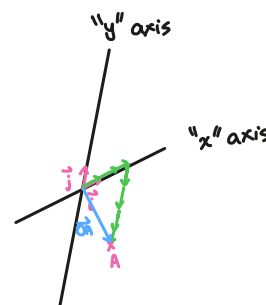
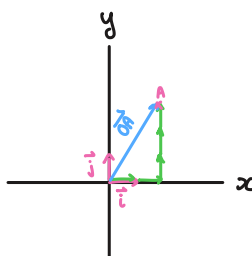
Any vector $\vec{OA}$ is expressed in its basis components:

$$\vec{OA} = 2\vec{i} + 3\vec{j}$$

where 2 and 3 are the **components** of $\vec{OA}$.

Non-standard basis ($\vec{i}'$, $\vec{j}'$ rotated/skewed): the *same* geometric vector gets different components, e.g.

$$\vec{OA} = 2.5\vec{i}' - 3.5\vec{j}'$$



Algebraic vectors have size and direction, but these are not obvious from the components alone.

Magnitude & Direction from Components

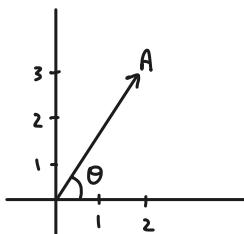
Given $\vec{OA} = 2\vec{i} + 3\vec{j}$:

Magnitude (Pythagorean theorem):

$$|\vec{OA}| = \sqrt{2^2 + 3^2} = \sqrt{13}$$

Direction (counterclockwise from positive x -axis):

$$\tan \theta = \frac{3}{2} \implies \theta = \tan^{-1}\left(\frac{3}{2}\right) \approx 56.3^\circ$$



Simplified & Column Vector Notation

When writing algebraic vectors using $\vec{i}$ and $\vec{j}$, we simplify by writing as an (x, y) point, but this represents a vector **starting at the origin** and **finishing at that point**:

$$\vec{OA} = 2\vec{i} + 3\vec{j} \longrightarrow \vec{OA} = (2, 3) = \begin{bmatrix} 2 \\ 3 \end{bmatrix}$$

The $\begin{bmatrix} x \\ y \end{bmatrix}$ form is called **column vector notation**.

6.2 Vector Addition

Recall: Geometric vectors are defined by their size and direction using systems such as [NESW] or simply drawing points on a page and connecting them.

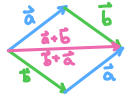
Algebraic Vectors have their size and direction defined by the Cartesian Plane (x, y) coordinates. The position of the tip (x, y) defines both the size and direction of the vector with the tail at the origin.

Vector addition is an operation we can perform on vectors that represents the result of two vectors positioned tip-to-tail.

Property: $\vec{a} + \vec{b} = \vec{b} + \vec{a}$ $\leftarrow$ Commutative

Proof:

"Vector proof"



Consider shape $ABCDEFGH$ (labelled parallelepiped):

Labelled parallelepiped $ABCDEFGH$ with vectors $\vec{a}$, $\vec{b}$ marked on edges

$$\overrightarrow{AB} + \overrightarrow{BF} = \overrightarrow{AF}$$

$$\overrightarrow{DB} + \overrightarrow{FH} = \vec{0}$$

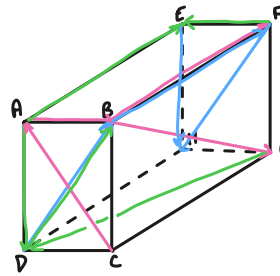
$$\overrightarrow{AD} + \overrightarrow{AE} = \overrightarrow{AH}$$

$$\overrightarrow{CA} + \overrightarrow{BG} = \overrightarrow{CA} + \overrightarrow{AH} = \overrightarrow{CH}$$

$$\overrightarrow{BF} - \overrightarrow{EH} = \overrightarrow{BF} + \overrightarrow{HE} = \overrightarrow{DH} + \overrightarrow{HE} = \overrightarrow{DE}$$

$$\overrightarrow{DB} + \overrightarrow{FE} + \overrightarrow{GD} = \overrightarrow{GD} + \overrightarrow{DB} + \overrightarrow{BA} = \overrightarrow{GA}$$

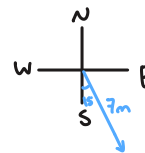
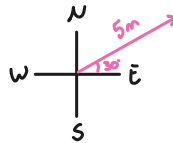
$$\overrightarrow{DB} + \overrightarrow{HF} = \overrightarrow{DB} + \overrightarrow{DB} = 2\overrightarrow{DB}$$



Example - Using [NESW] System

$$5 \text{ m [E } 30^\circ \text{ N]} + 7 \text{ m [S } 15^\circ \text{ E]} = \vec{c}$$

look E then turn 30° to the North
look S then turn 15° to the East



Adding vectors tip-to-tail will always produce a triangle. Use the **Cosine Law** and **Sine Law** to solve for the size and direction of $\vec{c}$.

Cos Law

$$c^2 = a^2 + b^2 - 2ab \cos C$$

$$c^2 = 5^2 + 7^2 - 2(5)(7) \cos 75^\circ$$

$$c = 7.48 \text{ m}$$

Sine law: $\frac{\sin A}{a} = \frac{\sin B}{b} = \frac{\sin C}{c}$

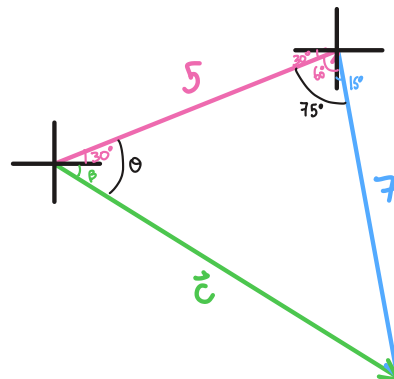
$$\frac{\sin \theta}{7} = \frac{\sin 75^\circ}{7.48}$$

$$\sin \theta = \frac{7 \sin 75^\circ}{7.48} = 0.90394$$

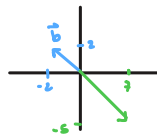
$$\theta = \sin^{-1}(0.90394) = 64.7^\circ$$

$$\beta = 64.7^\circ - 30^\circ = 34.7^\circ$$

$$\therefore c = 7.48 \text{ m [E } 34.7^\circ \text{ S]}$$

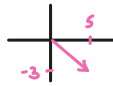


Adding Algebraic Vectors



Ex: $\vec{a} = (7, -5)$, $\vec{b} = (-2, 2)$

$$\vec{a} + \vec{b} = (7 + (-2), (-5) + 2) = (5, -3)$$



Find $|\vec{a} + \vec{b}|$:

$$|\vec{a} + \vec{b}| = \sqrt{(5)^2 + (-3)^2} = \sqrt{34}$$

* **Note:** $|\vec{a} + \vec{b}| \neq |\vec{a}| + |\vec{b}|$

$$|\vec{a}| = \sqrt{(7)^2 + (-5)^2} = \sqrt{74}$$

$$|\vec{b}| = \sqrt{(-2)^2 + (2)^2} = \sqrt{8}$$

$$\sqrt{74} + \sqrt{8} \neq \sqrt{34}$$

In general:

$$\begin{aligned} \vec{a} + \vec{b} &= (a_x, a_y) + (b_x, b_y) \\ &= (a_x + b_x, a_y + b_y) \end{aligned}$$

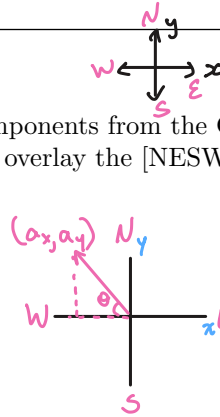
Algebraic Vectors - Geo/Alg Conversion

Algebraic vectors are written most commonly using the x and y components from the Cartesian grid. Geometric vectors have size and direction. We can convert from one to the other if we overlay the [NESW] system on the Cartesian grid.

Alg $\rightarrow$ Geo:

1. Let $\vec{a} = (a_x, a_y)$
2. Draw the vector with components to make a right triangle.
3. Find size $|\vec{a}|$ by Pythagorean Thm: $|\vec{a}| = \sqrt{(a_x)^2 + (a_y)^2}$
4. Find angle θ in triangle and state direction.

$$\tan \theta = \frac{a_y}{a_x} \Rightarrow \theta = ? \rightarrow [\text{W } \theta \text{ N}] \text{ for example}$$



Geo $\rightarrow$ Alg:

1. Use trig to find components x and y .
2. State the vector with appropriate "+" or "-" sign.

Ex: 5m[N30°W]

$$a_x = 5 \sin 30^\circ = 2.5[\text{W}], \quad a_y = 5 \cos 30^\circ = 2.5\sqrt{3}[\text{N}]$$

(* cos is close *)

$$\therefore \vec{a} = (-2.5, 2.5\sqrt{3})$$



Definition: Unit Vector

A unit vector has a magnitude of 1 in a certain direction. $(1, 0)$ is a unit vector and so is $(0, 1)$. $(1, 1)$ is not a unit vector because the magnitude of $(1, 1)$ is $\sqrt{1^2 + 1^2} = \sqrt{2}$ and the magnitude is obviously not 1.

Let $\vec{a}$ be any vector of any size in any direction. Then $\hat{a}$ is the unit vector in the direction of $\vec{a}$, and:

$$\hat{a} = \frac{1}{|\vec{a}|} \vec{a}$$

eg. $\frac{1}{\sqrt{2}}(1, 1) = \left(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right)$

$\uparrow$ $\uparrow$ $\uparrow$

$\frac{1}{|\vec{a}|}$ $\vec{a}$ unit vector $\hat{a}$

Ex: $\vec{a} = (3, 4)$ is not a unit vector since $|\vec{a}| = \sqrt{3^2 + 4^2} = 5$.

$$\hat{a} = \frac{1}{|\vec{a}|} \vec{a} = \frac{1}{5}(3, 4) = \left(\frac{3}{5}, \frac{4}{5}\right)$$

$\hat{a}$ is the unit vector in the direction of $\vec{a}$.

Check: $|\hat{a}| = \sqrt{\left(\frac{3}{5}\right)^2 + \left(\frac{4}{5}\right)^2} = \sqrt{\frac{9}{25} + \frac{16}{25}} = \sqrt{\frac{25}{25}} = \sqrt{1} = 1 \checkmark$

$\uparrow$ unit vector
in same direction
as $\vec{a}$

“Collinear”: $\hat{a} = k\vec{b}$, $\hat{a} = \frac{1}{5}\vec{a} \checkmark$

Vectors in Three Dimensions

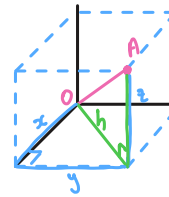
Three-Dimensional Vectors

$$\vec{a} = (a_x, a_y, a_z) \rightarrow \text{algebraic form}$$

A 3-D vector needs **3 components**.

Use the **3-D Pythagorean theorem** to find the magnitude:

$$|\vec{a}| = \sqrt{a_x^2 + a_y^2 + a_z^2}$$



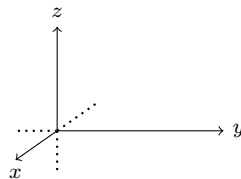
$$|\vec{OA}| = \sqrt{h^2 + z^2}$$

$$|\vec{OA}| = \sqrt{x^2 + y^2 + z^2}$$

$$h^2 = x^2 + y^2$$

Drawing a 3-D Point or Vector

To draw a vector or point in 3D, we need to use the perspective of a 3D box.

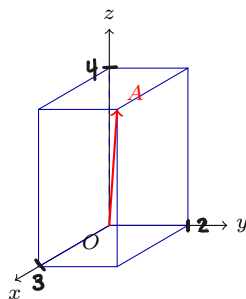


To draw a 3D point or vector:

1. Mark the x , y , and z values on their axis.
2. Draw a rectangle to join the y and z value.
3. Draw parallelograms to join x and z and x and y .
4. Complete the box with the missing corner (which is your 3D point).

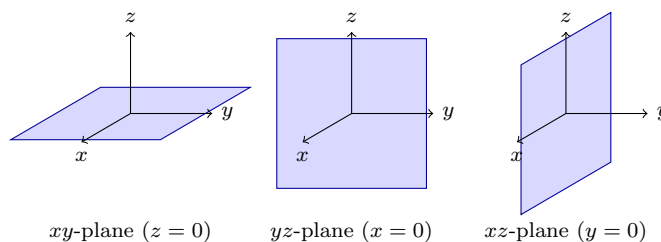
Octants and Coordinate Planes

Graph the point $A(3, 2, 4)$.



$$|\vec{OA}| = \sqrt{29}$$

The xyz grid has **8 octants**, and there are **3 basic planes** that separate those octants.



In 2-D:

$x = 0$ is the equation for the y -axis.

$y = 0$ is the equation for the x -axis.

$Ax + By + C = 0$ is the general equation of a line in $\mathbb{R}^2$.

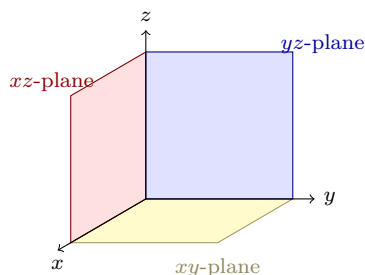
In 3-D:

$x = 0$ is the equation of the yz -plane (a flat surface in $\mathbb{R}^3$).

$y = 0$ is the equation of the xz -plane.

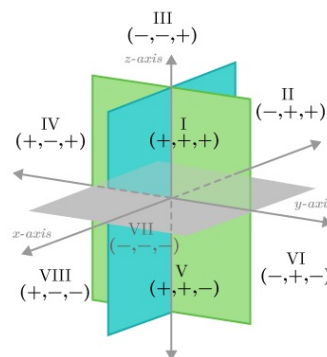
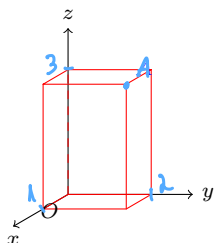
$z = 0$ is the equation of the xy -plane.

$Ax + By + Cz + D = 0$ is the general equation of a plane in $\mathbb{R}^3$.



Identifying Octants

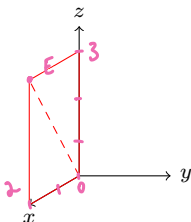
Example: Given $A(1, 2, 3)$ - which octant will this be in? - **Octant 1**



Octant	Sign conditions	Description
OCT 1	$x > 0, y > 0, z > 0$	Top, Front, Right
OCT 2	$x < 0, y > 0, z > 0$	Top, Back, Right
OCT 3	$x < 0, y < 0, z > 0$	Top, Back, Left
OCT 4	$x > 0, y < 0, z > 0$	Top, Front, Left
OCT 5	$x > 0, y > 0, z < 0$	Bottom, Front, Right
OCT 6	$x < 0, y > 0, z < 0$	Bottom, Back, Right
OCT 7	$x < 0, y < 0, z < 0$	Bottom, Back, Left
OCT 8	$x > 0, y < 0, z < 0$	Bottom, Front, Left

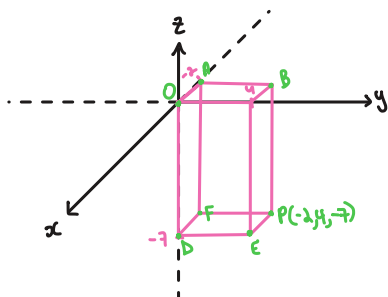
Position Vectors

Example: $E(2, 0, 3)$



Since E has $y = 0$, the point lies in the xz -plane - its “box” flattens to a rectangle.

Example: Coordinates of the vertices of the rectangular box shown:



$O(0, 0, 0)$

$A(-2, 0, 0)$

$B(-2, 4, 0)$

$C(0, 4, 0)$

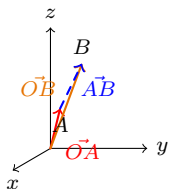
$D(0, 0, -7)$

$F(-2, 0, -7)$

$P(-2, 4, -7)$

$E(0, 4, -7)$

By graphing points in $\mathbb{R}^3$, we can locate the tip of a vector beginning at the origin. Such a vector is called a **position vector**.



$$\vec{AB} = \vec{OB} - \vec{OA}$$

This vector joining A to B is found using the position vectors $\vec{OA}$ and $\vec{OB}$ (point B minus point A).

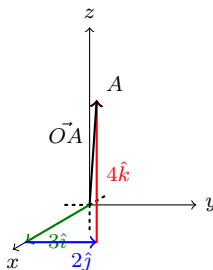
Standard Basis Vectors

Another way of writing algebraic vectors is to use the **standard basis vectors**. These are unit vectors along the x , y , and z axes.

$$\hat{i} = (1, 0, 0) \quad \hat{j} = (0, 1, 0) \quad \hat{k} = (0, 0, 1)$$

If $\vec{OA} = (3, 2, 4)$:

$$\vec{OA} = 3(1, 0, 0) + 2(0, 1, 0) + 4(0, 0, 1) = 3\hat{i} + 2\hat{j} + 4\hat{k}$$

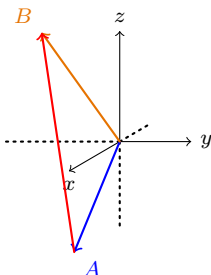


Recall: Use the 3D Pythagorean theorem to find the magnitude of a vector.

$$\text{If } \vec{a} = (a_x, a_y, a_z), \text{ then } |\vec{a}| = \sqrt{a_x^2 + a_y^2 + a_z^2}.$$

Eg: $A(4, 1, -2)$ and $B(-1, -3, 3)$. Find $\vec{AB}$ and $|\vec{AB}|$.

Worked Example



$$\begin{aligned} \vec{AB} &= \vec{OB} - \vec{OA} \\ &= (-1, -3, 3) - (4, 1, -2) \\ &= (-1 - 4, -3 - 1, 3 - (-2)) \\ \vec{AB} &= (-5, -4, 5) \\ |\vec{AB}| &= \sqrt{(-5)^2 + (-4)^2 + (5)^2} \\ &= \sqrt{66} \end{aligned}$$

Please note: $|\vec{AB}| \neq |\vec{OB}| - |\vec{OA}|$ — don't, just don't.

$$|\vec{OB}| = \sqrt{(-1)^2 + (-3)^2 + (3)^2} = \sqrt{19} \quad |\vec{OA}| = \sqrt{(4)^2 + (1)^2 + (-2)^2} = \sqrt{21}$$